

Formulas and Bounds for the Mostar Index of Kite Graphs and Their Inflations

Abstract

In this paper, we first determine the Mostar index of kite graphs and establish equalities and bounds related to this family of graphs. Next, we show that graph inflation increases the Mostar index, i.e.,

$$\text{Mo}(\text{inf}(\text{Kite}(p, q))) \geq \text{Mo}(\text{Kite}(p, q)),$$

and finally, we compute

$$\text{Mo}(\text{inf}(\text{Kite}(p, q))).$$

Role

I conducted a comprehensive literature review on the Mostar index and related graph families, identifying gaps in existing research. Using Python, I implemented various graph structures to facilitate hypothesis testing. I also created visualizations of these graphs to support the analysis and conclusions. Additionally, I proposed half of the novel hypotheses regarding the effects of inflation on the Mostar index, all of which have been proven, advancing our understanding of how these operations impact graph indices. We are currently revising the proofs and preparing to submit the paper to Communications in Combinatorics and Optimization (CCO).